

Machine Learning Project Ideas for Students

This document outlines five machine learning project ideas designed to reinforce concepts covered in Days 7-10 of an ML curriculum, including regression, classification, decision trees, random forests, clustering, cross-validation, and hyperparameter tuning. Each project provides a practical application of these techniques, encouraging students to engage with real-world datasets and develop a comprehensive understanding of the machine learning workflow.

Project 1: California Housing Price Prediction (Regression Focus)

Overview

This project focuses on building a regression model to predict median house values in various districts of California. Students will apply fundamental machine learning concepts, including data preprocessing, model training, and evaluation, using the well-known California Housing dataset.

Topics Covered

- * **ML Workflow**: Data loading, exploration, preprocessing, model training, evaluation, and deployment considerations.
 - * **Train/Test Split**: Dividing the dataset into training and testing sets to assess model generalization.
 - * **Linear Regression**: Implementing and understanding the principles of linear regression for continuous target variables.
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Dataset

The project utilizes the **California Housing dataset**, available through scikit-learn. This dataset contains 20,640 samples with 8 numerical features describing demographic and geographic characteristics of California districts. The target variable is the median house value, expressed in hundreds of thousands of dollars (\$100,000) [1].

Goal

The primary goal is to develop a linear regression model that accurately predicts the median house value for California districts based on the provided features.

Documentation and Explanation Points

- * **Feature Explanation**: Detail each feature in the dataset and its potential relevance to house prices.
 - * **Preprocessing (Scaling)**: Explain the necessity of feature scaling (e.g., StandardScaler) for linear regression and demonstrate its application.
 - * **Model Training**: Document the steps involved in training the linear regression model on the training data.
 - * **Evaluation Metrics**: Calculate and interpret key regression evaluation metrics such as Mean Squared Error (MSE) and R-squared (R²) on the test set.
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Project 2: Employee Attrition Prediction (Classification Focus)

Overview

This project addresses a common business problem: predicting employee attrition. Students will build a classification model using logistic regression to identify employees likely to leave the company, focusing on classification metrics and handling imbalanced datasets.

Topics Covered

- * **Logistic Regression**: Applying logistic regression for binary classification problems.
 - * **Classification Problems**: Understanding the nature of classification tasks and their unique challenges.
 - * **Evaluation Metrics**: In-depth analysis of precision, recall, F1-score, and their importance in classification.
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Dataset

The **IBM HR Analytics Employee Attrition & Performance dataset** from Kaggle will be used. This dataset contains information on approximately 1,470 employees with 35 variables related to demographics, job satisfaction, performance, and work conditions, with a target variable indicating attrition [2].

Goal

To develop a logistic regression model that can predict whether an employee will attrite (leave the company) or not, and to understand the factors contributing to attrition.

Documentation and Explanation Points

- * **Class Imbalance Handling**: Discuss strategies for addressing class imbalance (e.g., oversampling, undersampling, or using appropriate evaluation metrics).
 - * **Confusion Matrix**: Construct and interpret a confusion matrix to understand model performance across different classes.
 - * **ROC-AUC Curve**: Plot and explain the Receiver Operating Characteristic (ROC) curve and calculate the Area Under the Curve (AUC) as a robust evaluation metric for binary classifiers.
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Project 3: Heart Disease Risk Assessment (Tree-based Models)

Overview

This project involves classifying the risk of heart disease using decision trees and random forests. Students will explore the strengths of ensemble methods and learn how to extract feature importance to understand which factors are most indicative of heart disease.

Topics Covered

- * **Decision Trees**: Building and visualizing decision tree classifiers.
- * **Random Forest**: Implementing an ensemble of decision trees to improve prediction accuracy and robustness.

* **Feature Importance**: Identifying and interpreting the most influential features in the model.

Dataset

The **UCI Heart Disease Dataset** is suitable for this project. It contains various medical features such as age, sex, chest pain type, resting blood pressure, cholesterol levels, and a target variable indicating the presence of heart disease [3].

Goal

To build classification models (Decision Tree and Random Forest) to predict the presence of heart disease and to determine the most significant features contributing to the prediction.

Documentation and Explanation Points

- * **Visualizing Decision Trees**: Generate and interpret a visualization of a trained decision tree to understand its decision-making process.
 - * **Comparing Single Tree vs. Forest**: Analyze and compare the performance (accuracy, precision, recall) of a single decision tree against a random forest model.
 - * **Plotting Feature Importance**: Extract and visualize feature importance scores from the Random Forest model to highlight key risk factors for heart disease.
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Project 4: E-commerce Customer Segmentation (Clustering Focus)

Overview

This project applies unsupervised learning techniques, specifically K-Means clustering, to segment customers of an e-commerce platform. The goal is to identify distinct customer groups based on their purchasing behavior, which can inform targeted marketing strategies.

Topics Covered

- * **K-Means Clustering**: Understanding the K-Means algorithm and its application for grouping similar data points.

* **Customer Segmentation**: Applying clustering to categorize customers based on relevant features.

Dataset

A **Mall Customer Segmentation dataset** (commonly found on platforms like Kaggle) can be used. This dataset typically includes features such as customer ID, gender, age, annual income, and spending score [4].

Goal

To segment customers into meaningful groups based on their annual income and spending score, enabling businesses to tailor marketing campaigns to specific customer profiles.

Documentation and Explanation Points

- * **Elbow Method for Optimal K**: Explain and demonstrate the Elbow Method to determine the optimal number of clusters (K) for the K-Means algorithm.
 - * **Silhouette Score**: Calculate and interpret the Silhouette Score as another metric to evaluate the quality of the clustering.
 - * **Cluster Visualization**: Visualize the formed clusters using scatter plots, clearly distinguishing between different customer segments.
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Project 5: Credit Scoring Model Optimization (Advanced Tuning)

Overview

This project focuses on building and optimizing a credit scoring model, integrating concepts of cross-validation and hyperparameter tuning using Grid Search and Random Search. Students will learn to build a robust classification model and systematically improve its performance.

Topics Covered

- * **Cross Validation**: Implementing k-fold cross-validation to get a more reliable estimate of model performance.

- * **Grid Search**: Systematically searching through a predefined parameter space to find the best hyperparameters.
 - * **Random Search**: Randomly sampling hyperparameters from a distribution to efficiently explore the parameter space.
 - * **Project 2 Integration**: Applying these optimization techniques to a classification problem, potentially extending Project 2 or using a similar credit risk dataset.
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Dataset

A **Credit Scoring dataset** such as the German Credit Risk dataset or a similar dataset containing financial and demographic information for loan applicants, along with a target variable indicating creditworthiness, would be appropriate [5].

Goal

To develop a highly accurate and robust credit scoring model that predicts creditworthiness, and to optimize its performance using advanced hyperparameter tuning techniques.

Documentation and Explanation Points

- * **Comparison of Grid vs. Random Search**: Discuss the advantages and disadvantages of Grid Search and Random Search, and compare their effectiveness in finding optimal hyperparameters.
- * **5-fold Cross-Validation Results**: Present and interpret the results of 5-fold cross-validation for the chosen model, demonstrating its stability and generalization ability.
- * **Final Model Performance**: Report the performance of the optimized model using appropriate classification metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC) and compare it to a baseline model.